

## 1.1 BASIC AXIOMS AND EXAMPLES

8. Let  $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$ .
  - (a) Prove that  $G$  is a group under multiplication (called the group of *roots of unity* in  $\mathbb{C}$ ).
  - (b) Prove that  $G$  is not a group under addition.
9. Let  $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$ .
  - (a) Prove that  $G$  is a group under addition.
  - (b) Prove that the nonzero elements of  $G$  are a group under multiplication. [“Rationalize the denominators” to find multiplicative inverses.]

- 18.** Let  $x$  and  $y$  be elements of  $G$ . Prove that  $xy = yx$  if and only if  $y^{-1}xy = x$  if and only if  $x^{-1}y^{-1}xy = 1$ .
- 19.** Let  $x \in G$  and let  $a, b \in \mathbb{Z}^+$ .
- (a) Prove that  $x^{a+b} = x^a x^b$  and  $(x^a)^b = x^{ab}$ .
  - (b) Prove that  $(x^a)^{-1} = x^{-a}$ .
  - (c) Establish part (a) for arbitrary integers  $a$  and  $b$  (positive, negative or zero).
- 20.** For  $x$  an element in  $G$  show that  $x$  and  $x^{-1}$  have the same order.

21. Let  $G$  be a finite group and let  $x$  be an element of  $G$  of order  $n$ . Prove that if  $n$  is odd, then  $x = (x^2)^k$  for some  $k$ .
22. If  $x$  and  $g$  are elements of the group  $G$ , prove that  $|x| = |g^{-1}xg|$ . Deduce that  $|ab| = |ba|$  for all  $a, b \in G$ .
23. Suppose  $x \in G$  and  $|x| = n < \infty$ . If  $n = st$  for some positive integers  $s$  and  $t$ , prove that  $|x^s| = t$ .
24. If  $a$  and  $b$  are *commuting* elements of  $G$ , prove that  $(ab)^n = a^n b^n$  for all  $n \in \mathbb{Z}$ . [Do this by induction for positive  $n$  first.]
25. Prove that if  $x^2 = 1$  for all  $x \in G$  then  $G$  is abelian.

- 31.** Prove that any finite group  $G$  of even order contains an element of order 2. [Let  $t(G)$  be the set  $\{g \in G \mid g \neq g^{-1}\}$ . Show that  $t(G)$  has an even number of elements and every nonidentity element of  $G - t(G)$  has order 2.]
- 32.** If  $x$  is an element of finite order  $n$  in  $G$ , prove that the elements  $1, x, x^2, \dots, x^{n-1}$  are all distinct. Deduce that  $|x| \leq |G|$ .
- 33.** Let  $x$  be an element of finite order  $n$  in  $G$ .
- (a) Prove that if  $n$  is odd then  $x^i \neq x^{-i}$  for all  $i = 1, 2, \dots, n-1$ .
  - (b) Prove that if  $n = 2k$  and  $1 \leq i < n$  then  $x^i = x^{-i}$  if and only if  $i = k$ .

- 35.** If  $x$  is an element of finite order  $n$  in  $G$ , use the Division Algorithm to show that *any* integral power of  $x$  equals one of the elements in the set  $\{1, x, x^2, \dots, x^{n-1}\}$  (so these are all the distinct elements of the cyclic subgroup (cf. Exercise 27 above) of  $G$  generated by  $x$ ).
- 36.** Assume  $G = \{1, a, b, c\}$  is a group of order 4 with identity 1. Assume also that  $G$  has no elements of order 4 (so by Exercise 32, every element has order  $\leq 3$ ). Use the cancellation laws to show that there is a unique group table for  $G$ . Deduce that  $G$  is abelian.

## 1.2 DIHEDRAL GROUPS

2. Use the generators and relations above to show that if  $x$  is any element of  $D_{2n}$  which is not a power of  $r$ , then  $rx = xr^{-1}$ .
3. Use the generators and relations above to show that every element of  $D_{2n}$  which is not a power of  $r$  has order 2. Deduce that  $D_{2n}$  is generated by the two elements  $s$  and  $sr$ , both of which have order 2.

4. If  $n = 2k$  is even and  $n \geq 4$ , show that  $z = r^k$  is an element of order 2 which commutes with all elements of  $D_{2n}$ . Show also that  $z$  is the only nonidentity element of  $D_{2n}$  which commutes with all elements of  $D_{2n}$ . [cf. Exercise 33 of Section 1.]
5. If  $n$  is odd and  $n \geq 3$ , show that the identity is the only element of  $D_{2n}$  which commutes with all elements of  $D_{2n}$ . [cf. Exercise 33 of Section 1.]
6. Let  $x$  and  $y$  be elements of order 2 in any group  $G$ . Prove that if  $t = xy$  then  $tx = xt^{-1}$  (so that if  $n = |xy| < \infty$  then  $x, t$  satisfy the same relations in  $G$  as  $s, r$  do in  $D_{2n}$ ).

7. Show that  $\langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$  gives a presentation for  $D_{2n}$  in terms of the two generators  $a = s$  and  $b = sr$  of order 2 computed in Exercise 3 above. [Show that the relations for  $r$  and  $s$  follow from the relations for  $a$  and  $b$  and, conversely, the relations for  $a$  and  $b$  follow from those for  $r$  and  $s$ .]
8. Find the order of the cyclic subgroup of  $D_{2n}$  generated by  $r$  (cf. Exercise 27 of Section 1).



- 16.** Show that the group  $\langle x_1, y_1 \mid x_1^2 = y_1^2 = (x_1 y_1)^2 = 1 \rangle$  is the dihedral group  $D_4$  (where  $x_1$  may be replaced by the letter  $r$  and  $y_1$  by  $s$ ). [Show that the last relation is the same as:  $x_1 y_1 = y_1 x_1^{-1}$ .]
- 17.** Let  $X_{2n}$  be the group whose presentation is displayed in (1.2).
- (a) Show that if  $n = 3k$ , then  $X_{2n}$  has order 6, and it has the same generators and relations as  $D_6$  when  $x$  is replaced by  $r$  and  $y$  by  $s$ .
  - (b) Show that if  $(3, n) = 1$ , then  $x$  satisfies the additional relation:  $x = 1$ . In this case deduce that  $X_{2n}$  has order 2. [Use the facts that  $x^n = 1$  and  $x^3 = 1$ .]

**18.** Let  $Y$  be the group whose presentation is displayed in (1.3).

- (a) Show that  $v^2 = v^{-1}$ . [Use the relation:  $v^3 = 1$ .]
- (b) Show that  $v$  commutes with  $u^3$ . [Show that  $v^2 u^3 v = u^3$  by writing the left hand side as  $(v^2 u^2)(uv)$  and using the relations to reduce this to the right hand side. Then use part (a).]
- (c) Show that  $v$  commutes with  $u$ . [Show that  $u^9 = u$  and then use part (b).]
- (d) Show that  $uv = 1$ . [Use part (c) and the last relation.]
- (e) Show that  $u = 1$ , deduce that  $v = 1$ , and conclude that  $Y = 1$ . [Use part (d) and the equation  $u^4 v^3 = 1$ .]

10. Prove that if  $\sigma$  is the  $m$ -cycle  $(a_1 a_2 \dots a_m)$ , then for all  $i \in \{1, 2, \dots, m\}$ ,  $\sigma^i(a_k) = a_{k+i}$ , where  $k+i$  is replaced by its least residue mod  $m$  when  $k+i > m$ . Deduce that  $|\sigma| = m$ .
11. Let  $\sigma$  be the  $m$ -cycle  $(1 2 \dots m)$ . Show that  $\sigma^i$  is also an  $m$ -cycle if and only if  $i$  is relatively prime to  $m$ .
12. (a) If  $\tau = (1 2)(3 4)(5 6)(7 8)(9 10)$  determine whether there is a  $n$ -cycle  $\sigma$  ( $n \geq 10$ ) with  $\tau = \sigma^k$  for some integer  $k$ .  
 (b) If  $\tau = (1 2)(3 4 5)$  determine whether there is an  $n$ -cycle  $\sigma$  ( $n \geq 5$ ) with  $\tau = \sigma^k$  for some integer  $k$ .
13. Show that an element has order 2 in  $S_n$  if and only if its cycle decomposition is a product of commuting 2-cycles.

10. Since  $\sigma(a_k) = a_{k+1}$ , clearly by the induction,

11.  $\sigma^i$  is  $m$ -cycle iff  $1, \sigma^i(1), \sigma^{2i}(1), \dots, \sigma^{(m-1)i}(1)$  are distinct.  
 iff  $1, 1+i, 1+2i, \dots, 1+(m-1)i$  are distinct up to mod  $m$ .  
 iff for  $0 \leq x, y \leq m-1$ ,  $x \neq y \Rightarrow x i \not\equiv y i \pmod{m}$   
 iff  $ai \equiv 0 \pmod{m}$ ,  $1 \leq a \leq m-1$   
 iff  $(m, i) = 1$ .  
 (if  $(m, i) = d > 1$ , then  
 for  $a = \frac{m}{d}$ ,  $\frac{m}{d} i$  is divided by  $m$ .

- 14.** Let  $p$  be a prime. Show that an element has order  $p$  in  $S_n$  if and only if its cycle decomposition is a product of commuting  $p$ -cycles. Show by an explicit example that this need not be the case if  $p$  is not prime.
- 15.** Prove that the order of an element in  $S_n$  equals the least common multiple of the lengths of the cycles in its cycle decomposition. [Use Exercise 10 and Exercise 24 of Section 1.]
- 16.** Show that if  $n \geq m$  then the number of  $m$ -cycles in  $S_n$  is given by

$$\frac{n(n-1)(n-2)\cdots(n-m+1)}{m}.$$

[Count the number of ways of forming an  $m$ -cycle and divide by the number of representations of a particular  $m$ -cycle.]

- 17.** Show that if  $n \geq 4$  then the number of permutations in  $S_n$  which are the product of two disjoint 2-cycles is  $n(n-1)(n-2)(n-3)/8$ .
- 18.** Find all numbers  $n$  such that  $S_5$  contains an element of order  $n$ . [Use Exercise 15.]
- 19.** Find all numbers  $n$  such that  $S_7$  contains an element of order  $n$ . [Use Exercise 15.]
- 20.** Find a set of generators and relations for  $S_3$ .

## 1.6 HOMOMORPHISMS AND ISOMORPHISMS

Let  $G$  and  $H$  be groups.

1. Let  $\varphi : G \rightarrow H$  be a homomorphism.
  - (a) Prove that  $\varphi(x^n) = \varphi(x)^n$  for all  $n \in \mathbb{Z}^+$ .
  - (b) Do part (a) for  $n = -1$  and deduce that  $\varphi(x^n) = \varphi(x)^n$  for all  $n \in \mathbb{Z}$ .
2. If  $\varphi : G \rightarrow H$  is an isomorphism, prove that  $|\varphi(x)| = |x|$  for all  $x \in G$ . Deduce that any two isomorphic groups have the same number of elements of order  $n$  for each  $n \in \mathbb{Z}^+$ . Is the result true if  $\varphi$  is only assumed to be a homomorphism?
3. If  $\varphi : G \rightarrow H$  is an isomorphism, prove that  $G$  is abelian if and only if  $H$  is abelian. If  $\varphi : G \rightarrow H$  is a homomorphism, what additional conditions on  $\varphi$  (if any) are sufficient to ensure that if  $G$  is abelian, then so is  $H$ ?

7. Prove that  $D_8$  and  $Q_8$  are not isomorphic.
8. Prove that if  $n \neq m$ ,  $S_n$  and  $S_m$  are not isomorphic.
9. Prove that  $D_{24}$  and  $S_4$  are not isomorphic.

11. Let  $A$  and  $B$  be groups. Prove that  $A \times B \cong B \times A$ .
12. Let  $A$ ,  $B$ , and  $C$  be groups and let  $G = A \times B$  and  $H = B \times C$ . Prove that  $G \times C \cong A \times H$ .
13. Let  $G$  and  $H$  be groups and let  $\varphi : G \rightarrow H$  be a homomorphism. Prove that the image of  $\varphi$ ,  $\varphi(G)$ , is a subgroup of  $H$  (cf. Exercise 26 of Section 1). Prove that if  $\varphi$  is injective then  $G \cong \varphi(G)$ .
14. Let  $G$  and  $H$  be groups and let  $\varphi : G \rightarrow H$  be a homomorphism. Define the *kernel* of  $\varphi$  to be  $\{g \in G \mid \varphi(g) = 1_H\}$  (so the kernel is the set of elements in  $G$  which map to the identity of  $H$ , i.e., is the fiber over the identity of  $H$ ). Prove that the kernel of  $\varphi$  is a subgroup (cf. Exercise 26 of Section 1) of  $G$ . Prove that  $\varphi$  is injective if and only if the kernel of  $\varphi$  is the identity subgroup of  $G$ .



- 23.** Let  $G$  be a finite group which possesses an automorphism  $\sigma$  (cf. Exercise 20) such that  $\sigma(g) = g$  if and only if  $g = 1$ . If  $\sigma^2$  is the identity map from  $G$  to  $G$ , prove that  $G$  is abelian (such an automorphism  $\sigma$  is called *fixed point free* of order 2). [Show that every element of  $G$  can be written in the form  $x^{-1}\sigma(x)$  and apply  $\sigma$  to such an expression.]
- 24.** Let  $G$  be a finite group and let  $x$  and  $y$  be distinct elements of order 2 in  $G$  that generate  $G$ . Prove that  $G \cong D_{2n}$ , where  $n = |xy|$ . [See Exercise 6 in Section 2.]

**25.** Let  $n \in \mathbb{Z}^+$ , let  $r$  and  $s$  be the usual generators of  $D_{2n}$  and let  $\theta = 2\pi/n$ .

(a) Prove that the matrix  $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$  is the matrix of the linear transformation which rotates the  $x, y$  plane about the origin in a counterclockwise direction by  $\theta$  radians.

(b) Prove that the map  $\varphi : D_{2n} \rightarrow GL_2(\mathbb{R})$  defined on generators by

$$\varphi(r) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad \text{and} \quad \varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism of  $D_{2n}$  into  $GL_2(\mathbb{R})$ .

(c) Prove that the homomorphism  $\varphi$  in part (b) is injective.

**26.** Let  $i$  and  $j$  be the generators of  $Q_8$  described in Section 5. Prove that the map  $\varphi$  from  $Q_8$  to  $GL_2(\mathbb{C})$  defined on generators by

$$\varphi(i) = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \quad \text{and} \quad \varphi(j) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism. Prove that  $\varphi$  is injective.

## 1.7 GROUP ACTIONS

4. Let  $G$  be a group acting on a set  $A$  and fix some  $a \in A$ . Show that the following sets are subgroups of  $G$  (cf. Exercise 26 of Section 1):
  - (a) the kernel of the action,
  - (b)  $\{g \in G \mid ga = a\}$  — this subgroup is called the *stabilizer* of  $a$  in  $G$ .
5. Prove that the kernel of an action of the group  $G$  on the set  $A$  is the same as the kernel of the corresponding permutation representation  $G \rightarrow S_A$  (cf. Exercise 14 in Section 6).
6. Prove that a group  $G$  acts faithfully on a set  $A$  if and only if the kernel of the action is the set consisting only of the identity.

8. Let  $A$  be a nonempty set and let  $k$  be a positive integer with  $k \leq |A|$ . The symmetric group  $S_A$  acts on the set  $B$  consisting of all subsets of  $A$  of cardinality  $k$  by  $\sigma \cdot \{a_1, \dots, a_k\} = \{\sigma(a_1), \dots, \sigma(a_k)\}$ .
- (a) Prove that this is a group action.
  - (b) Describe explicitly how the elements  $(1\ 2)$  and  $(1\ 2\ 3)$  act on the six 2-element subsets of  $\{1, 2, 3, 4\}$ .

14. Let  $G$  be a group and let  $A = G$ . Show that if  $G$  is non-abelian then the maps defined by  $g \cdot a = ag$  for all  $g, a \in G$  do *not* satisfy the axioms of a (left) group action of  $G$  on itself.
15. Let  $G$  be any group and let  $A = G$ . Show that the maps defined by  $g \cdot a = ag^{-1}$  for all  $g, a \in G$  *do* satisfy the axioms of a (left) group action of  $G$  on itself.
16. Let  $G$  be any group and let  $A = G$ . Show that the maps defined by  $g \cdot a = gag^{-1}$  for all  $g, a \in G$  *do* satisfy the axioms of a (left) group action (this action of  $G$  on itself is called *conjugation*).
17. Let  $G$  be a group and let  $G$  act on itself by left conjugation, so each  $g \in G$  maps  $G$  to  $G$  by

$$x \mapsto gxg^{-1}.$$

For fixed  $g \in G$ , prove that conjugation by  $g$  is an isomorphism from  $G$  onto itself (i.e., is an automorphism of  $G$  — cf. Exercise 20, Section 6). Deduce that  $x$  and  $gxg^{-1}$  have the same order for all  $x$  in  $G$  and that for any subset  $A$  of  $G$ ,  $|A| = |gAg^{-1}|$  (here  $gAg^{-1} = \{gag^{-1} \mid a \in A\}$ ).

**18.** Let  $H$  be a group acting on a set  $A$ . Prove that the relation  $\sim$  on  $A$  defined by

$$a \sim b \quad \text{if and only if} \quad a = hb \quad \text{for some } h \in H$$

is an equivalence relation. (For each  $x \in A$  the equivalence class of  $x$  under  $\sim$  is called the *orbit* of  $x$  under the action of  $H$ . The orbits under the action of  $H$  partition the set  $A$ .)

**19.** Let  $H$  be a subgroup (cf. Exercise 26 of Section 1) of the finite group  $G$  and let  $H$  act on  $G$  (here  $A = G$ ) by left multiplication. Let  $x \in G$  and let  $\mathcal{O}$  be the orbit of  $x$  under the action of  $H$ . Prove that the map

$$H \rightarrow \mathcal{O} \quad \text{defined by} \quad h \mapsto hx$$

is a bijection (hence all orbits have cardinality  $|H|$ ). From this and the preceding exercise deduce *Lagrange's Theorem*:

*if  $G$  is a finite group and  $H$  is a subgroup of  $G$  then  $|H|$  divides  $|G|$ .*

# Subgroups

## 2.2 CENTRALIZERS AND NORMALIZERS, STABILIZERS AND KERNELS

1. Prove that  $C_G(A) = \{g \in G \mid g^{-1}ag = a \text{ for all } a \in A\}$ .
2. Prove that  $C_G(Z(G)) = G$  and deduce that  $N_G(Z(G)) = G$ .
3. Prove that if  $A$  and  $B$  are subsets of  $G$  with  $A \subseteq B$  then  $C_G(B)$  is a subgroup of  $C_G(A)$ .

6. Let  $H$  be a subgroup of the group  $G$ .

- (a) Show that  $H \leq N_G(H)$ . Give an example to show that this is not necessarily true if  $H$  is not a subgroup.
- (b) Show that  $H \leq C_G(H)$  if and only if  $H$  is abelian.



5. In each of parts (a) to (c) show that for the specified group  $G$  and subgroup  $A$  of  $G$ ,  $C_G(A) = A$  and  $N_G(A) = G$ .
- (a)  $G = S_3$  and  $A = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$ .
  - (b)  $G = D_8$  and  $A = \{1, s, r^2, sr^2\}$ .
  - (c)  $G = D_{10}$  and  $A = \{1, r, r^2, r^3, r^4\}$ .
7. Let  $n \in \mathbb{Z}$  with  $n \geq 3$ . Prove the following:
- (a)  $Z(D_{2n}) = 1$  if  $n$  is odd
  - (b)  $Z(D_{2n}) = \{1, r^k\}$  if  $n = 2k$ .

8. Let  $G = S_n$ , fix an  $i \in \{1, 2, \dots, n\}$  and let  $G_i = \{\sigma \in G \mid \sigma(i) = i\}$  (the stabilizer of  $i$  in  $G$ ). Use group actions to prove that  $G_i$  is a subgroup of  $G$ . Find  $|G_i|$ .
9. For any subgroup  $H$  of  $G$  and any nonempty subset  $A$  of  $G$  define  $N_H(A)$  to be the set  $\{h \in H \mid hAh^{-1} = A\}$ . Show that  $N_H(A) = N_G(A) \cap H$  and deduce that  $N_H(A)$  is a subgroup of  $H$  (note that  $A$  need not be a subset of  $H$ ).
10. Let  $H$  be a subgroup of order 2 in  $G$ . Show that  $N_G(H) = C_G(H)$ . Deduce that if  $N_G(H) = G$  then  $H \leq Z(G)$ .
11. Prove that  $Z(G) \leq N_G(A)$  for any subset  $A$  of  $G$ .

- 13.** Let  $n$  be a positive integer and let  $R$  be the set of all polynomials with integer coefficients in the independent variables  $x_1, x_2, \dots, x_n$ , i.e., the members of  $R$  are finite sums of elements of the form  $ax_1^{r_1}x_2^{r_2}\cdots x_n^{r_n}$ , where  $a$  is any integer and  $r_1, \dots, r_n$  are nonnegative integers. For each  $\sigma \in S_n$  define a map

$$\sigma : R \rightarrow R \quad \text{by} \quad \sigma \cdot p(x_1, x_2, \dots, x_n) = p(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(n)}).$$

Prove that this defines a (left) group action of  $S_n$  on  $R$ .

## 2.3 CYCLIC GROUPS AND CYCLIC SUBGROUPS

16. Assume  $|x| = n$  and  $|y| = m$ . Suppose that  $x$  and  $y$  *commute*:  $xy = yx$ . Prove that  $|xy|$  divides the least common multiple of  $m$  and  $n$ . Need this be true if  $x$  and  $y$  do *not* commute? Give an example of commuting elements  $x, y$  such that the order of  $xy$  is not equal to the least common multiple of  $|x|$  and  $|y|$ .
17. Find a presentation for  $Z_n$  with one generator.
18. Show that if  $H$  is any group and  $h$  is an element of  $H$  with  $h^n = 1$ , then there is a unique homomorphism from  $Z_n = \langle x \rangle$  to  $H$  such that  $x \mapsto h$ .
19. Show that if  $H$  is any group and  $h$  is an element of  $H$ , then there is a unique homomorphism from  $\mathbb{Z}$  to  $H$  such that  $1 \mapsto h$ .

- 20.** Let  $p$  be a prime and let  $n$  be a positive integer. Show that if  $x$  is an element of the group  $G$  such that  $x^{p^n} = 1$  then  $|x| = p^m$  for some  $m \leq n$ .
- 21.** Let  $p$  be an odd prime and let  $n$  be a positive integer. Use the Binomial Theorem to show that  $(1 + p)^{p^{n-1}} \equiv 1 \pmod{p^n}$  but  $(1 + p)^{p^{n-2}} \not\equiv 1 \pmod{p^n}$ . Deduce that  $1 + p$  is an element of order  $p^{n-1}$  in the multiplicative group  $(\mathbb{Z}/p^n\mathbb{Z})^\times$ .
- 22.** Let  $n$  be an integer  $\geq 3$ . Use the Binomial Theorem to show that  $(1 + 2^2)^{2^{n-2}} \equiv 1 \pmod{2^n}$  but  $(1 + 2^2)^{2^{n-3}} \not\equiv 1 \pmod{2^n}$ . Deduce that 5 is an element of order  $2^{n-2}$  in the multiplicative group  $(\mathbb{Z}/2^n\mathbb{Z})^\times$ .
- 23.** Show that  $(\mathbb{Z}/2^n\mathbb{Z})^\times$  is not cyclic for any  $n \geq 3$ . [Find two distinct subgroups of order 2.]

**24.** Let  $G$  be a finite group and let  $x \in G$ .

(a) Prove that if  $g \in N_G(\langle x \rangle)$  then  $gxg^{-1} = x^a$  for some  $a \in \mathbb{Z}$ .

(b) Prove conversely that if  $gxg^{-1} = x^a$  for some  $a \in \mathbb{Z}$  then  $g \in N_G(\langle x \rangle)$ . [Show first that  $gx^k g^{-1} = (gxg^{-1})^k = x^{ak}$  for any integer  $k$ , so that  $g\langle x \rangle g^{-1} \leq \langle x \rangle$ . If  $x$  has order  $n$ , show the elements  $gx^i g^{-1}$ ,  $i = 0, 1, \dots, n-1$  are distinct, so that  $|g\langle x \rangle g^{-1}| = |\langle x \rangle| = n$  and conclude that  $g\langle x \rangle g^{-1} = \langle x \rangle$ .]

Note that this cuts down some of the work in computing normalizers of cyclic subgroups since one does not have to check  $ghg^{-1} \in \langle x \rangle$  for every  $h \in \langle x \rangle$ .

**25.** Let  $G$  be a cyclic group of order  $n$  and let  $k$  be an integer relatively prime to  $n$ . Prove that the map  $x \mapsto x^k$  is surjective. Use Lagrange's Theorem (Exercise 19, Section 1.7) to prove the same is true for any finite group of order  $n$ . (For such  $k$  each element has a  $k^{\text{th}}$  root in  $G$ . It follows from Cauchy's Theorem in Section 3.2 that if  $k$  is not relatively prime to the order of  $G$  then the map  $x \mapsto x^k$  is not surjective.)

**26.** Let  $Z_n$  be a cyclic group of order  $n$  and for each integer  $a$  let

$$\sigma_a : Z_n \rightarrow Z_n \quad \text{by} \quad \sigma_a(x) = x^a \text{ for all } x \in Z_n.$$

- (a) Prove that  $\sigma_a$  is an automorphism of  $Z_n$  if and only if  $a$  and  $n$  are relatively prime (automorphisms were introduced in Exercise 20, Section 1.6).
- (b) Prove that  $\sigma_a = \sigma_b$  if and only if  $a \equiv b \pmod{n}$ .
- (c) Prove that *every* automorphism of  $Z_n$  is equal to  $\sigma_a$  for some integer  $a$ .
- (d) Prove that  $\sigma_a \circ \sigma_b = \sigma_{ab}$ . Deduce that the map  $\bar{a} \mapsto \sigma_a$  is an isomorphism of  $(\mathbb{Z}/n\mathbb{Z})^\times$  onto the automorphism group of  $Z_n$  (so  $\text{Aut}(Z_n)$  is an abelian group of order  $\varphi(n)$ ).

## 2.4 SUBGROUPS GENERATED BY SUBSETS OF A GROUP

1. Prove that if  $H$  is a subgroup of  $G$  then  $\langle H \rangle = H$ .
2. Prove that if  $A$  is a subset of  $B$  then  $\langle A \rangle \leq \langle B \rangle$ . Give an example where  $A \subseteq B$  with  $A \neq B$  but  $\langle A \rangle = \langle B \rangle$ .
3. Prove that if  $H$  is an abelian subgroup of a group  $G$  then  $\langle H, Z(G) \rangle$  is abelian. Give an explicit example of an abelian subgroup  $H$  of a group  $G$  such that  $\langle H, C_G(H) \rangle$  is not abelian.
4. Prove that if  $H$  is a subgroup of  $G$  then  $H$  is generated by the set  $H - \{1\}$ .

1. By the definition,  $\langle H \rangle = \bigcap_{\substack{H \subseteq S \\ S \leq G}} S = H$ .

2.  $\langle B \rangle = \bigcap_{\substack{B \subseteq S \\ S \leq G}} S \supseteq \bigcap_{\substack{A \subseteq S \\ S \leq G}} S = \langle A \rangle$ . More precisely, let  $x \in \langle A \rangle$ .

Then, for all  $S \leq G$  s.t.  $A \subseteq S$ ,  $x \in S$ . And,  $A \subseteq B$  implies that  $S \leq G$  and  $B \subseteq S \Rightarrow x \in S$ , thus  $x \in \langle B \rangle$ .

Example:  $G = \mathbb{Z}_5$ ,  $A = \{1\}$ ,  $B = \{1, 2\}$ . (Since  $\mathbb{Z}_5$  has no proper non-trivial subgroup, directly.)

3. Let  $x, y \in \langle H, Z(G) \rangle$  be given. Then we can write

$$x = hk, y = pg$$

where  $h, p \in H$  and  $k, g \in Z(G)$ . Now,

$$xy = hkpg = \underset{\substack{\uparrow \\ Z(G)}}{h} \underset{\substack{\uparrow \\ H}}{p} \underset{\substack{\uparrow \\ Z(G)}}{k} g = pg hk$$

Thus abelian.

Example:  $G = D_4$ ,  $H = \langle r \rangle$ . Since  $C_G(\langle r \rangle) = \langle r \rangle$ ,  $\langle \langle r \rangle, C_G(\langle r \rangle) \rangle = \langle r \rangle \neq G$ .

4.  $\langle H \setminus \{1\} \rangle \leq \langle H \rangle = H$  is clear. Conversely, for any  $x \in H$ ,  $x = 1 \Rightarrow x \in \langle H \setminus \{1\} \rangle$ ,  
 $x \neq 1 \Rightarrow x \in H \setminus \{1\} \subseteq \langle H \setminus \{1\} \rangle$ .



6. Prove that the subgroup of  $S_4$  generated by  $(1\ 2)$  and  $(1\ 2)(3\ 4)$  is a noncyclic group of order 4.
7. Prove that the subgroup of  $S_4$  generated by  $(1\ 2)$  and  $(1\ 3)(2\ 4)$  is isomorphic to the dihedral group of order 8.
8. Prove that  $S_4 = \langle (1\ 2\ 3\ 4), (1\ 2\ 4\ 3) \rangle$ .

6. observe that:  $(1\ 2)^2 = 1$ ,  $((1\ 2)(3\ 4))^2 = 1$ ,  $(1\ 2)((1\ 2)(3\ 4)) = (3\ 4)$ ,  
 $(1\ 2)(3\ 4)(1\ 2) = (3\ 4)$ ,  $(3\ 4)^2 = 1$

therefore  $\langle (1\ 2), (1\ 2)(3\ 4) \rangle = \{1, (1\ 2), (3\ 4), (1\ 2)(3\ 4)\}$ , And all element have order less than 2,  
 thus non-cyclic.

---

17.

9. Prove that  $SL_2(\mathbb{F}_3)$  is the subgroup of  $GL_2(\mathbb{F}_3)$  generated by  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ . [Recall from Exercise 9 of Section 1 that  $SL_2(\mathbb{F}_3)$  is the subgroup of matrices of determinant 1. You may assume this subgroup has order 24 — this will be an exercise in Section 3.2.]
10. Prove that the subgroup of  $SL_2(\mathbb{F}_3)$  generated by  $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  is isomorphic to the quaternion group of order 8. [Use a presentation for  $Q_8$ .]
11. Show that  $SL_2(\mathbb{F}_3)$  and  $S_4$  are two nonisomorphic groups of order 24.
12. Prove that the subgroup of upper triangular matrices in  $GL_3(\mathbb{F}_2)$  is isomorphic to the dihedral group of order 8 (cf. Exercise 16, Section 1). [First find the order of this subgroup.]

14. A group  $H$  is called *finitely generated* if there is a finite set  $A$  such that  $H = \langle A \rangle$ .
- (a) Prove that every finite group is finitely generated.
  - (b) Prove that  $\mathbb{Z}$  is finitely generated.
  - (c) Prove that every finitely generated subgroup of the additive group  $\mathbb{Q}$  is cyclic. [If  $H$  is a finitely generated subgroup of  $\mathbb{Q}$ , show that  $H \leq \langle \frac{1}{k} \rangle$ , where  $k$  is the product of all the denominators which appear in a set of generators for  $H$ .]
  - (d) Prove that  $\mathbb{Q}$  is not finitely generated.
15. Exhibit a proper subgroup of  $\mathbb{Q}$  which is not cyclic.

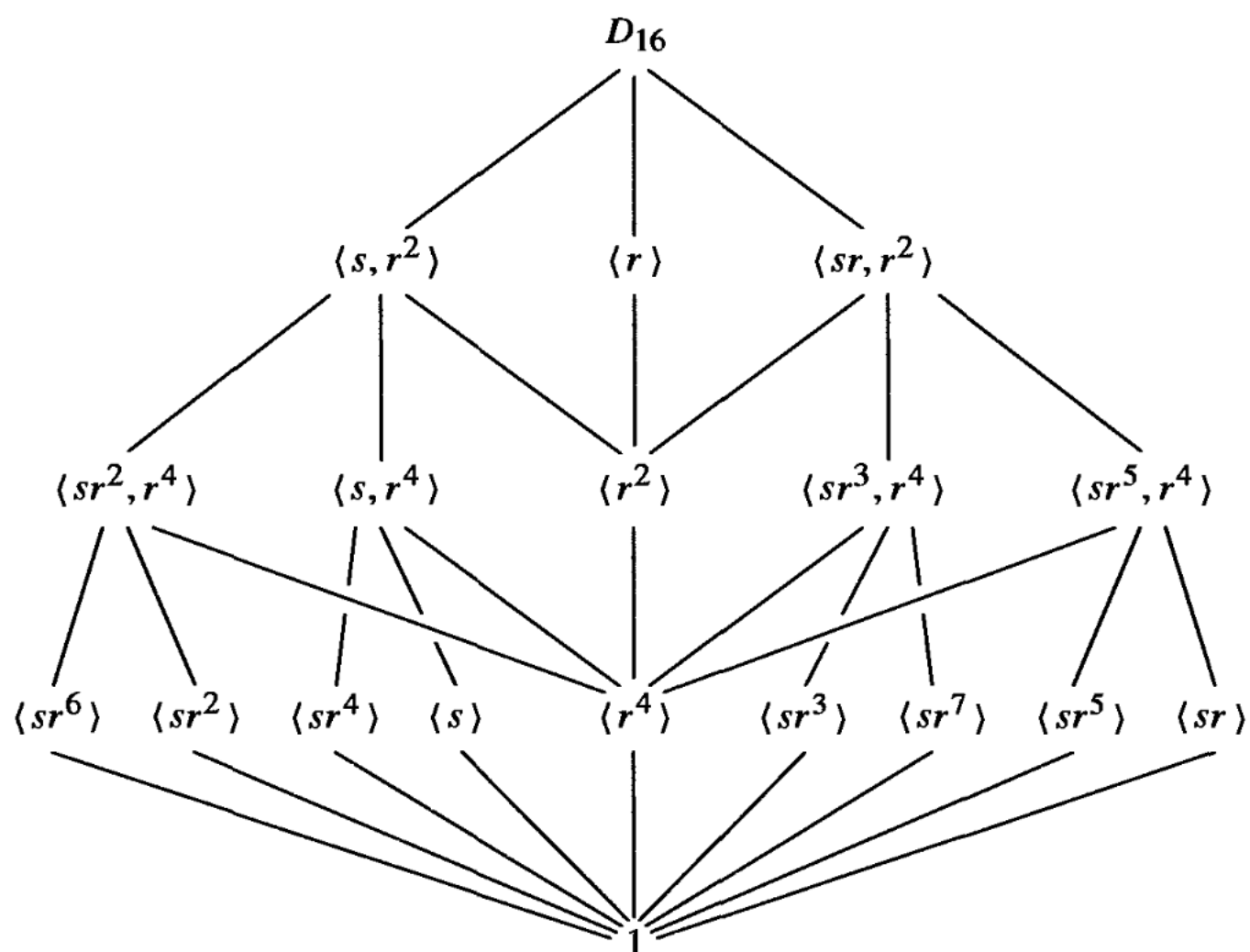
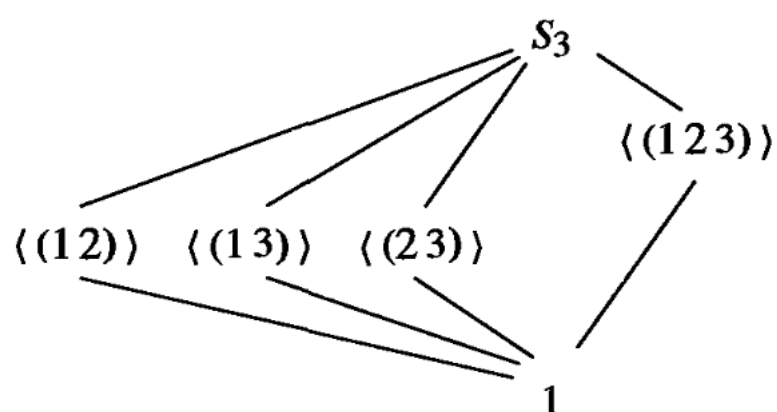
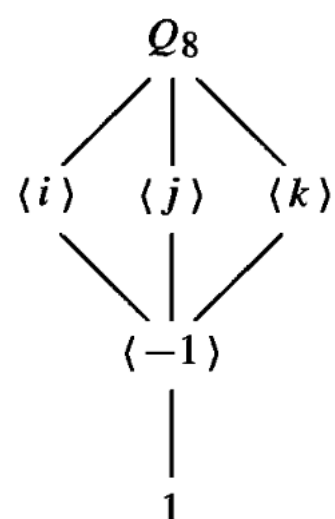
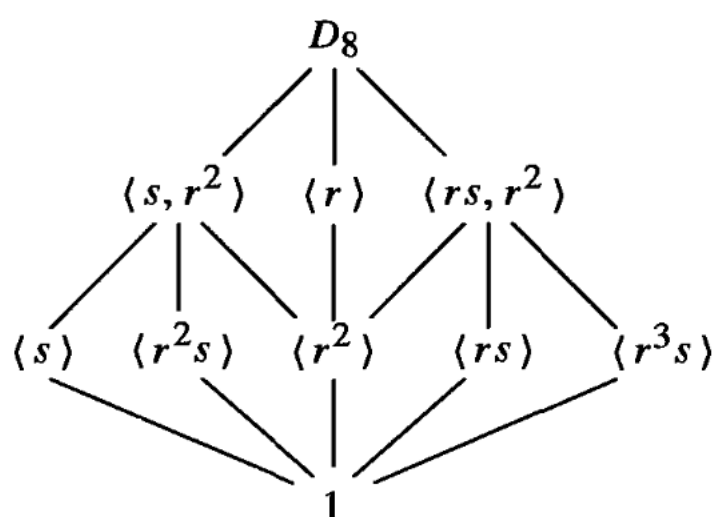
- 16.** A subgroup  $M$  of a group  $G$  is called a *maximal subgroup* if  $M \neq G$  and the only subgroups of  $G$  which contain  $M$  are  $M$  and  $G$ .
- (a) Prove that if  $H$  is a proper subgroup of the finite group  $G$  then there is a maximal subgroup of  $G$  containing  $H$ .
  - (b) Show that the subgroup of all rotations in a dihedral group is a maximal subgroup.
  - (c) Show that if  $G = \langle x \rangle$  is a cyclic group of order  $n \geq 1$  then a subgroup  $H$  is maximal if and only if  $H = \langle x^p \rangle$  for some prime  $p$  dividing  $n$ .

- 17.** This is an exercise involving Zorn's Lemma (see Appendix I) to prove that every nontrivial finitely generated group possesses maximal subgroups. Let  $G$  be a finitely generated group, say  $G = \langle g_1, g_2, \dots, g_n \rangle$ , and let  $\mathcal{S}$  be the set of all proper subgroups of  $G$ . Then  $\mathcal{S}$  is partially ordered by inclusion. Let  $\mathcal{C}$  be a chain in  $\mathcal{S}$ .
- (a) Prove that the union,  $H$ , of all the subgroups in  $\mathcal{C}$  is a subgroup of  $G$ .
  - (b) Prove that  $H$  is a *proper* subgroup. [If not, each  $g_i$  must lie in  $H$  and so must lie in some element of the chain  $\mathcal{C}$ . Use the definition of a chain to arrive at a contradiction.]
  - (c) Use Zorn's Lemma to show that  $\mathcal{S}$  has a maximal element (which is, by definition, a maximal subgroup).

18. Let  $p$  be a prime and let  $Z = \{z \in \mathbb{C} \mid z^{p^n} = 1 \text{ for some } n \in \mathbb{Z}^+\}$  (so  $Z$  is the multiplicative group of all  $p$ -power roots of unity in  $\mathbb{C}$ ). For each  $k \in \mathbb{Z}^+$  let  $H_k = \{z \in Z \mid z^{p^k} = 1\}$  (the group of  $p^k$ th roots of unity). Prove the following:
- (a)  $H_k \leq H_m$  if and only if  $k \leq m$
  - (b)  $H_k$  is cyclic for all  $k$  (assume that for any  $n \in \mathbb{Z}^+$ ,  $\{e^{2\pi it/n} \mid t = 0, 1, \dots, n-1\}$  is the set of all  $n^{\text{th}}$  roots of 1 in  $\mathbb{C}$ )
  - (c) every proper subgroup of  $Z$  equals  $H_k$  for some  $k \in \mathbb{Z}^+$  (in particular, every proper subgroup of  $Z$  is finite and cyclic)
  - (d)  $Z$  is not finitely generated.

- 19.** A nontrivial abelian group  $A$  (written multiplicatively) is called *divisible* if for each element  $a \in A$  and each nonzero integer  $k$  there is an element  $x \in A$  such that  $x^k = a$ , i.e., each element has a  $k^{\text{th}}$  root in  $A$  (in additive notation, each element is the  $k^{\text{th}}$  multiple of some element of  $A$ ).
- (a) Prove that the additive group of rational numbers,  $\mathbb{Q}$ , is divisible.
  - (b) Prove that no finite abelian group is divisible.
- 20.** Prove that if  $A$  and  $B$  are nontrivial abelian groups, then  $A \times B$  is divisible if and only if both  $A$  and  $B$  are divisible groups.

## 2.5 THE LATTICE OF SUBGROUPS OF A GROUP





1. Let  $H$  and  $K$  be subgroups of  $G$ . Exhibit all possible sublattices which show only  $G$ ,  $1$ ,  $H$ ,  $K$  and their joins and intersections. What distinguishes the different drawings?
2. In each of (a) to (d) list all subgroups of  $D_{16}$  that satisfy the given condition.
  - (a) Subgroups that are contained in  $\langle sr^2, r^4 \rangle$
  - (b) Subgroups that are contained in  $\langle sr^7, r^4 \rangle$
  - (c) Subgroups that contain  $\langle r^4 \rangle$
  - (d) Subgroups that contain  $\langle s \rangle$ .
3. Show that the subgroup  $\langle s, r^2 \rangle$  of  $D_8$  is isomorphic to  $V_4$ .

7. Find the center of  $D_{16}$ .
8. In each of the following groups find the normalizer of each subgroup:  
(a)  $S_3$       (b)  $Q_8$ .
9. Draw the lattices of subgroups of the following groups:  
(a)  $\mathbb{Z}/16\mathbb{Z}$       (b)  $\mathbb{Z}/24\mathbb{Z}$       (c)  $\mathbb{Z}/48\mathbb{Z}$ . [See Exercise 6 in Section 3.]
10. Classify groups of order 4 by proving that if  $|G| = 4$  then  $G \cong \mathbb{Z}_4$  or  $G \cong V_4$ . [See Exercise 36, Section 1.1.]

**11.** Consider the group of order 16 with the following presentation:

$$QD_{16} = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma\tau = \tau\sigma^3 \rangle$$

(called the *quasidihedral* or *semidihedral* group of order 16). This group has three subgroups of order 8:  $\langle \tau, \sigma^2 \rangle \cong D_8$ ,  $\langle \sigma \rangle \cong Z_8$  and  $\langle \sigma^2, \sigma\tau \rangle \cong Q_8$  and every proper subgroup is contained in one of these three subgroups. Fill in the missing subgroups in the lattice of all subgroups of the quasidihedral group on the following page, exhibiting each subgroup with at most two generators. (This is another example of a nonplanar lattice.)

- 12.** The group  $A = Z_2 \times Z_4 = \langle a, b \mid a^2 = b^4 = 1, ab = ba \rangle$  has order 8 and has three subgroups of order 4:  $\langle a, b^2 \rangle \cong V_4$ ,  $\langle b \rangle \cong Z_4$  and  $\langle ab \rangle \cong Z_4$  and every proper subgroup is contained in one of these three. Draw the lattice of all subgroups of  $A$ , giving each subgroup in terms of at most two generators.
- 13.** The group  $G = Z_2 \times Z_8 = \langle x, y \mid x^2 = y^8 = 1, xy = yx \rangle$  has order 16 and has three subgroups of order 8:  $\langle x, y^2 \rangle \cong Z_2 \times Z_4$ ,  $\langle y \rangle \cong Z_8$  and  $\langle xy \rangle \cong Z_8$  and every proper subgroup is contained in one of these three. Draw the lattice of all subgroups of  $G$ , giving each subgroup in terms of at most two generators (cf. Exercise 12).

**14.** Let  $M$  be the group of order 16 with the following presentation:

$$\langle u, v \mid u^2 = v^8 = 1, vu = uv^5 \rangle$$

(sometimes called the *modular* group of order 16). It has three subgroups of order 8:  $\langle u, v^2 \rangle$ ,  $\langle v \rangle$  and  $\langle uv \rangle$  and every proper subgroup is contained in one of these three. Prove that  $\langle u, v^2 \rangle \cong Z_2 \times Z_4$ ,  $\langle v \rangle \cong Z_8$  and  $\langle uv \rangle \cong Z_8$ . Show that the lattice of subgroups of  $M$  is the same as the lattice of subgroups of  $Z_2 \times Z_8$  (cf. Exercise 13) but that these two groups are not isomorphic.

16. Use the lattice of subgroups of the quasidihedral group of order 16 to show that every element of order 2 is contained in the proper subgroup  $\langle \tau, \sigma^2 \rangle$  (cf. Exercise 11).
17. Use the lattice of subgroups of the modular group  $M$  of order 16 to show that the set  $\{x \in M \mid x^2 = 1\}$  is a subgroup of  $M$  isomorphic to the Klein 4-group (cf. Exercise 14).
18. Use the lattice to help find the centralizer of every element of  $QD_{16}$  (cf. Exercise 11).
19. Use the lattice to help find  $N_{D_{16}}(\langle s, r^4 \rangle)$ .
20. Use the lattice of subgroups of  $QD_{16}$  (cf. Exercise 11) to help find the normalizers
  - (a)  $N_{QD_{16}}(\langle \tau \sigma \rangle)$
  - (b)  $N_{QD_{16}}(\langle \tau, \sigma^4 \rangle)$ .

# Quotient Groups and Homomorphisms

1. Let  $\varphi : G \rightarrow H$  be a homomorphism and let  $E$  be a subgroup of  $H$ . Prove that  $\varphi^{-1}(E) \leq G$  (i.e., the preimage or pullback of a subgroup under a homomorphism is a subgroup). If  $E \trianglelefteq H$  prove that  $\varphi^{-1}(E) \trianglelefteq G$ . Deduce that  $\ker \varphi \trianglelefteq G$ .
2. Let  $\varphi : G \rightarrow H$  be a homomorphism of groups with kernel  $K$  and let  $a, b \in \varphi(G)$ . Let  $X \in G/K$  be the fiber above  $a$  and let  $Y$  be the fiber above  $b$ , i.e.,  $X = \varphi^{-1}(a)$ ,  $Y = \varphi^{-1}(b)$ . Fix an element  $u$  of  $X$  (so  $\varphi(u) = a$ ). Prove that if  $XY = Z$  in the quotient group  $G/K$  and  $w$  is any member of  $Z$ , then there is some  $v \in Y$  such that  $uv = w$ . [Show  $u^{-1}w \in Y$ .]

3. Let  $A$  be an abelian group and let  $B$  be a subgroup of  $A$ . Prove that  $A/B$  is abelian. Give an example of a non-abelian group  $G$  containing a proper normal subgroup  $N$  such that  $G/N$  is abelian.
4. Prove that in the quotient group  $G/N$ ,  $(gN)^\alpha = g^\alpha N$  for all  $\alpha \in \mathbb{Z}$ .
5. Use the preceding exercise to prove that the order of the element  $gN$  in  $G/N$  is  $n$ , where  $n$  is the smallest positive integer such that  $g^n \in N$  (and  $gN$  has infinite order if no such positive integer exists). Give an example to show that the order of  $gN$  in  $G/N$  may be strictly smaller than the order of  $g$  in  $G$ .



**11.** Let  $F$  be a field and let  $G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in F, ac \neq 0 \right\} \leq GL_2(F)$ .

(a) Prove that the map  $\varphi : \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mapsto a$  is a surjective homomorphism from  $G$  onto  $F^\times$  (recall that  $F^\times$  is the multiplicative group of nonzero elements in  $F$ ). Describe the fibers and kernel of  $\varphi$ .

(b) Prove that the map  $\psi : \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mapsto (a, c)$  is a surjective homomorphism from  $G$  onto  $F^\times \times F^\times$ . Describe the fibers and kernel of  $\psi$ .

(c) Let  $H = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in F \right\}$ . Prove that  $H$  is isomorphic to the additive group  $F$ .

- 16.** Let  $G$  be a group, let  $N$  be a normal subgroup of  $G$  and let  $\overline{G} = G/N$ . Prove that if  $G = \langle x, y \rangle$  then  $\overline{G} = \langle \overline{x}, \overline{y} \rangle$ . Prove more generally that if  $G = \langle S \rangle$  for any subset  $S$  of  $G$ , then  $\overline{G} = \langle \overline{S} \rangle$ .

**17.** Let  $G$  be the dihedral group of order 16 (whose lattice appears in Section 2.5):

$$G = \langle r, s \mid r^8 = s^2 = 1, rs = sr^{-1} \rangle$$

and let  $\overline{G} = G/\langle r^4 \rangle$  be the quotient of  $G$  by the subgroup generated by  $r^4$  (this subgroup is the center of  $G$ , hence is normal).

- (a) Show that the order of  $\overline{G}$  is 8.
- (b) Exhibit each element of  $\overline{G}$  in the form  $\overline{s}^a \overline{r}^b$ , for some integers  $a$  and  $b$ .
- (c) Find the order of each of the elements of  $\overline{G}$  exhibited in (b).
- (d) Write each of the following elements of  $\overline{G}$  in the form  $\overline{s}^a \overline{r}^b$ , for some integers  $a$  and  $b$  as in (b):  $\overline{rs}$ ,  $\overline{sr^{-2}s}$ ,  $\overline{s^{-1}r^{-1}sr}$ .
- (e) Prove that  $\overline{H} = \langle \overline{s}, \overline{r}^2 \rangle$  is a normal subgroup of  $\overline{G}$  and  $\overline{H}$  is isomorphic to the Klein 4-group. Describe the isomorphism type of the complete preimage of  $\overline{H}$  in  $G$ .
- (f) Find the center of  $\overline{G}$  and describe the isomorphism type of  $\overline{G}/Z(\overline{G})$ .

- 18.** Let  $G$  be the quasidihedral group of order 16 (whose lattice was computed in Exercise 11 of Section 2.5):

$$G = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma\tau = \tau\sigma^3 \rangle$$

and let  $\overline{G} = G/\langle \sigma^4 \rangle$  be the quotient of  $G$  by the subgroup generated by  $\sigma^4$  (this subgroup is the center of  $G$ , hence is normal).

- (a) Show that the order of  $\overline{G}$  is 8.
- (b) Exhibit each element of  $\overline{G}$  in the form  $\overline{\tau^a \sigma^b}$ , for some integers  $a$  and  $b$ .
- (c) Find the order of each of the elements of  $\overline{G}$  exhibited in (b).
- (d) Write each of the following elements of  $\overline{G}$  in the form  $\overline{\tau^a \sigma^b}$ , for some integers  $a$  and  $b$  as in (b):  $\overline{\sigma\tau}$ ,  $\overline{\tau\sigma^{-2}\tau}$ ,  $\overline{\tau^{-1}\sigma^{-1}\tau\sigma}$ .
- (e) Prove that  $\overline{G} \cong D_8$ .

19. Let  $G$  be the modular group of order 16 (whose lattice was computed in Exercise 14 of Section 2.5):

$$G = \langle u, v \mid u^2 = v^8 = 1, vu = uv^5 \rangle$$

and let  $\overline{G} = G/\langle v^4 \rangle$  be the quotient of  $G$  by the subgroup generated by  $v^4$  (this subgroup is contained in the center of  $G$ , hence is normal).

- (a) Show that the order of  $\overline{G}$  is 8.
- (b) Exhibit each element of  $\overline{G}$  in the form  $\overline{u}^a \overline{v}^b$ , for some integers  $a$  and  $b$ .
- (c) Find the order of each of the elements of  $\overline{G}$  exhibited in (b).
- (d) Write each of the following elements of  $\overline{G}$  in the form  $\overline{u}^a \overline{v}^b$ , for some integers  $a$  and  $b$  as in (b):  $\overline{vu}$ ,  $\overline{uv^{-2}u}$ ,  $\overline{u^{-1}v^{-1}uv}$ .
- (e) Prove that  $\overline{G}$  is abelian and is isomorphic to  $Z_2 \times Z_4$ .

**21.** Let  $G = Z_4 \times Z_4$  be given in terms of the following generators and relations:

$$G = \langle x, y \mid x^4 = y^4 = 1, xy = yx \rangle.$$

Let  $\overline{G} = G/\langle x^2y^2 \rangle$  (note that every subgroup of the abelian group  $G$  is normal).

- (a) Show that the order of  $\overline{G}$  is 8.
- (b) Exhibit each element of  $\overline{G}$  in the form  $\overline{x}^a\overline{y}^b$ , for some integers  $a$  and  $b$ .
- (c) Find the order of each of the elements of  $\overline{G}$  exhibited in (b).
- (d) Prove that  $\overline{G} \cong Z_4 \times Z_2$ .

- 22. (a)** Prove that if  $H$  and  $K$  are normal subgroups of a group  $G$  then their intersection  $H \cap K$  is also a normal subgroup of  $G$ .
- (b)** Prove that the intersection of an arbitrary nonempty collection of normal subgroups of a group is a normal subgroup (do not assume the collection is countable).
- 23.** Prove that the join (cf. Section 2.5) of any nonempty collection of normal subgroups of a group is a normal subgroup.
- 24.** Prove that if  $N \trianglelefteq G$  and  $H$  is any subgroup of  $G$  then  $N \cap H \trianglelefteq H$ .

25. (a) Prove that a subgroup  $N$  of  $G$  is normal if and only if  $gNg^{-1} \subseteq N$  for *all*  $g \in G$ .  
(b) Let  $G = GL_2(\mathbb{Q})$ , let  $N$  be the subgroup of upper triangular matrices with integer entries and 1's on the diagonal, and let  $g$  be the diagonal matrix with entries 2,1. Show that  $gNg^{-1} \subseteq N$  but  $g$  does *not* normalize  $N$ .
26. Let  $a, b \in G$ .  
(a) Prove that the conjugate of the product of  $a$  and  $b$  is the product of the conjugate of  $a$  and the conjugate of  $b$ . Prove that the order of  $a$  and the order of any conjugate of  $a$  are the same.  
(b) Prove that the conjugate of  $a^{-1}$  is the inverse of the conjugate of  $a$ .  
(c) Let  $N = \langle S \rangle$  for some subset  $S$  of  $G$ . Prove that  $N \trianglelefteq G$  if  $gSg^{-1} \subseteq N$  for all  $g \in G$ .  
(d) Deduce that if  $N$  is the cyclic group  $\langle x \rangle$ , then  $N$  is normal in  $G$  if and only if for each  $g \in G$ ,  $gxg^{-1} = x^k$  for some  $k \in \mathbb{Z}$ .  
(e) Let  $n$  be a positive integer. Prove that the subgroup  $N$  of  $G$  generated by all the elements of  $G$  of order  $n$  is a normal subgroup of  $G$ .



27. Let  $N$  be a *finite* subgroup of a group  $G$ . Show that  $gNg^{-1} \subseteq N$  if and only if  $gNg^{-1} = N$ . Deduce that  $N_G(N) = \{g \in G \mid gNg^{-1} \subseteq N\}$ .
28. Let  $N$  be a *finite* subgroup of a group  $G$  and assume  $N = \langle S \rangle$  for some subset  $S$  of  $G$ . Prove that an element  $g \in G$  normalizes  $N$  if and only if  $gSg^{-1} \subseteq N$ .
29. Let  $N$  be a *finite* subgroup of  $G$  and suppose  $G = \langle T \rangle$  and  $N = \langle S \rangle$  for some subsets  $S$  and  $T$  of  $G$ . Prove that  $N$  is normal in  $G$  if and only if  $tSt^{-1} \subseteq N$  for all  $t \in T$ .
30. Let  $N \leq G$  and let  $g \in G$ . Prove that  $gN = Ng$  if and only if  $g \in N_G(N)$ .
31. Prove that if  $H \leq G$  and  $N$  is a normal subgroup of  $H$  then  $H \leq N_G(N)$ . Deduce that  $N_G(N)$  is the largest subgroup of  $G$  in which  $N$  is normal (i.e., is the join of all subgroups  $H$  for which  $N \trianglelefteq H$ ).

- 34.** Let  $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$  be the usual presentation of the dihedral group of order  $2n$  and let  $k$  be a positive integer dividing  $n$ .
- (a) Prove that  $\langle r^k \rangle$  is a normal subgroup of  $D_{2n}$ .
  - (b) Prove that  $D_{2n}/\langle r^k \rangle \cong D_{2k}$ .
- 36.** Prove that if  $G/Z(G)$  is cyclic then  $G$  is abelian. [If  $G/Z(G)$  is cyclic with generator  $xZ(G)$ , show that every element of  $G$  can be written in the form  $x^a z$  for some integer  $a \in \mathbb{Z}$  and some element  $z \in Z(G)$ .]

37. Let  $A$  and  $B$  be groups. Show that  $\{(a, 1) \mid a \in A\}$  is a normal subgroup of  $A \times B$  and the quotient of  $A \times B$  by this subgroup is isomorphic to  $B$ .
38. Let  $A$  be an abelian group and let  $D$  be the (diagonal) subgroup  $\{(a, a) \mid a \in A\}$  of  $A \times A$ . Prove that  $D$  is a normal subgroup of  $A \times A$  and  $(A \times A)/D \cong A$ .
39. Suppose  $A$  is the non-abelian group  $S_3$  and  $D$  is the diagonal subgroup  $\{(a, a) \mid a \in A\}$  of  $A \times A$ . Prove that  $D$  is not normal in  $A \times A$ .

40. Let  $G$  be a group, let  $N$  be a normal subgroup of  $G$  and let  $\overline{G} = G/N$ . Prove that  $\overline{x}$  and  $\overline{y}$  commute in  $\overline{G}$  if and only if  $x^{-1}y^{-1}xy \in N$ . (The element  $x^{-1}y^{-1}xy$  is called the *commutator* of  $x$  and  $y$  and is denoted by  $[x, y]$ .)
41. Let  $G$  be a group. Prove that  $N = \langle x^{-1}y^{-1}xy \mid x, y \in G \rangle$  is a normal subgroup of  $G$  and  $G/N$  is abelian ( $N$  is called the *commutator subgroup* of  $G$ ).
42. Assume both  $H$  and  $K$  are normal subgroups of  $G$  with  $H \cap K = 1$ . Prove that  $xy = yx$  for all  $x \in H$  and  $y \in K$ . [Show  $x^{-1}y^{-1}xy \in H \cap K$ .]

### 3.2 MORE ON COSETS AND LAGRANGE'S THEOREM

4. Show that if  $|G| = pq$  for some primes  $p$  and  $q$  (not necessarily distinct) then either  $G$  is abelian or  $Z(G) = 1$ . [See Exercise 36 in Section 1.]
5. Let  $H$  be a subgroup of  $G$  and fix some element  $g \in G$ .
  - (a) Prove that  $gHg^{-1}$  is a subgroup of  $G$  of the same order as  $H$ .
  - (b) Deduce that if  $n \in \mathbb{Z}^+$  and  $H$  is the unique subgroup of  $G$  of order  $n$  then  $H \trianglelefteq G$ .
6. Let  $H \leq G$  and let  $g \in G$ . Prove that if the right coset  $Hg$  equals *some* left coset of  $H$  in  $G$  then it equals the left coset  $gH$  and  $g$  must be in  $N_G(H)$ .
7. Let  $H \leq G$  and define a relation  $\sim$  on  $G$  by  $a \sim b$  if and only if  $b^{-1}a \in H$ . Prove that  $\sim$  is an equivalence relation and describe the equivalence class of each  $a \in G$ . Use this to prove Proposition 4.
8. Prove that if  $H$  and  $K$  are finite subgroups of  $G$  whose orders are relatively prime then  $H \cap K = 1$ .

- 10.** Suppose  $H$  and  $K$  are subgroups of finite index in the (possibly infinite) group  $G$  with  $|G : H| = m$  and  $|G : K| = n$ . Prove that  $\text{l.c.m.}(m, n) \leq |G : H \cap K| \leq mn$ . Deduce that if  $m$  and  $n$  are relatively prime then  $|G : H \cap K| = |G : H| \cdot |G : K|$ .
- 11.** Let  $H \leq K \leq G$ . Prove that  $|G : H| = |G : K| \cdot |K : H|$  (do not assume  $G$  is finite).
- 12.** Let  $H \leq G$ . Prove that the map  $x \mapsto x^{-1}$  sends each left coset of  $H$  in  $G$  onto a right coset of  $H$  and gives a bijection between the set of left cosets and the set of right cosets of  $H$  in  $G$  (hence the number of left cosets of  $H$  in  $G$  equals the number of right cosets).

- 14.** Prove that  $S_4$  does not have a normal subgroup of order 8 or a normal subgroup of order 3.
- 15.** Let  $G = S_n$  and for fixed  $i \in \{1, 2, \dots, n\}$  let  $G_i$  be the stabilizer of  $i$ . Prove that  $G_i \cong S_{n-1}$ .

- 16.** Use Lagrange's Theorem in the multiplicative group  $(\mathbb{Z}/p\mathbb{Z})^\times$  to prove *Fermat's Little Theorem*: if  $p$  is a prime then  $a^p \equiv a \pmod{p}$  for all  $a \in \mathbb{Z}$ .
- 17.** Let  $p$  be a prime and let  $n$  be a positive integer. Find the order of  $\bar{p}$  in  $(\mathbb{Z}/(p^n-1)\mathbb{Z})^\times$  and deduce that  $n \mid \varphi(p^n - 1)$  (here  $\varphi$  is Euler's function).
- 22.** Use Lagrange's Theorem in the multiplicative group  $(\mathbb{Z}/n\mathbb{Z})^\times$  to prove *Euler's Theorem*:  $a^{\varphi(n)} \equiv 1 \pmod{n}$  for every integer  $a$  relatively prime to  $n$ , where  $\varphi$  denotes Euler's  $\varphi$ -function.



18. Let  $G$  be a finite group, let  $H$  be a subgroup of  $G$  and let  $N \trianglelefteq G$ . Prove that if  $|H|$  and  $|G : N|$  are relatively prime then  $H \leq N$ .
19. Prove that if  $N$  is a normal subgroup of the finite group  $G$  and  $(|N|, |G : N|) = 1$  then  $N$  is the unique subgroup of  $G$  of order  $|N|$ .
20. If  $A$  is an abelian group with  $A \trianglelefteq G$  and  $B$  is any subgroup of  $G$  prove that  $A \cap B \trianglelefteq AB$ .

### 3.3 THE ISOMORPHISM THEOREMS

2. Prove all parts of the Lattice Isomorphism Theorem.
3. Prove that if  $H$  is a normal subgroup of  $G$  of prime index  $p$  then for all  $K \leq G$  either
  - (i)  $K \leq H$  or
  - (ii)  $G = HK$  and  $|K : K \cap H| = p$ .
4. Let  $C$  be a normal subgroup of the group  $A$  and let  $D$  be a normal subgroup of the group  $B$ . Prove that  $(C \times D) \trianglelefteq (A \times B)$  and  $(A \times B)/(C \times D) \cong (A/C) \times (B/D)$ .

7. Let  $M$  and  $N$  be normal subgroups of  $G$  such that  $G = MN$ . Prove that  $G/(M \cap N) \cong (G/M) \times (G/N)$ . [Draw the lattice.]
8. Let  $p$  be a prime and let  $G$  be the group of  $p$ -power roots of 1 in  $\mathbb{C}$  (cf. Exercise 18, Section 2.4). Prove that the map  $z \mapsto z^p$  is a surjective homomorphism. Deduce that  $G$  is isomorphic to a proper quotient of itself.
9. Let  $p$  be a prime and let  $G$  be a group of order  $p^a m$ , where  $p$  does not divide  $m$ . Assume  $P$  is a subgroup of  $G$  of order  $p^a$  and  $N$  is a normal subgroup of  $G$  of order  $p^b n$ , where  $p$  does not divide  $n$ . Prove that  $|P \cap N| = p^b$  and  $|PN/N| = p^{a-b}$ . (The subgroup  $P$  of  $G$  is called a *Sylow  $p$ -subgroup* of  $G$ . This exercise shows that the intersection of any Sylow  $p$ -subgroup of  $G$  with a normal subgroup  $N$  is a Sylow  $p$ -subgroup of  $N$ .)
10. Generalize the preceding exercise as follows. A subgroup  $H$  of a finite group  $G$  is called a *Hall subgroup* of  $G$  if its index in  $G$  is relatively prime to its order:  $(|G : H|, |H|) = 1$ . Prove that if  $H$  is a Hall subgroup of  $G$  and  $N \trianglelefteq G$ , then  $H \cap N$  is a Hall subgroup of  $N$  and  $HN/N$  is a Hall subgroup of  $G/N$ .

### 3.5 TRANSPOSITIONS AND THE ALTERNATING GROUP

2. Prove that  $\sigma^2$  is an even permutation for every permutation  $\sigma$ .
3. Prove that  $S_n$  is generated by  $\{(i \ i+1) \mid 1 \leq i \leq n-1\}$ . [Consider conjugates, viz.  $(2\ 3)(1\ 2)(2\ 3)^{-1}$ .]
4. Show that  $S_n = \langle (1\ 2), (1\ 2\ 3 \dots n) \rangle$  for all  $n \geq 2$ .
5. Show that if  $p$  is prime,  $S_p = \langle \sigma, \tau \rangle$  where  $\sigma$  is any transposition and  $\tau$  is any  $p$ -cycle.

6. Show that  $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$  is a proper subgroup of  $S_4$ . What is the isomorphism type of this subgroup?
7. Prove that the group of rigid motions of a tetrahedron is isomorphic to  $A_4$ . [Recall Exercise 20 in Section 1.7.]
8. Prove the lattice of subgroups of  $A_4$  given in the text is correct. [By the preceding exercise and the comments following Lagrange's Theorem,  $A_4$  has no subgroup of order 6.]
9. Prove that the (unique) subgroup of order 4 in  $A_4$  is normal and is isomorphic to  $V_4$ .
10. Find a composition series for  $A_4$ . Deduce that  $A_4$  is solvable.
11. Prove that  $S_4$  has no subgroup isomorphic to  $Q_8$ .

12. Prove that  $A_n$  contains a subgroup isomorphic to  $S_{n-2}$  for each  $n \geq 3$ .
13. Prove that every element of order 2 in  $A_n$  is the square of an element of order 4 in  $S_n$ . [An element of order 2 in  $A_n$  is a product of  $2k$  commuting transpositions.]
14. Prove that the subgroup of  $A_4$  generated by any element of order 2 and any element of order 3 is all of  $A_4$ .
15. Prove that if  $x$  and  $y$  are distinct 3-cycles in  $S_4$  with  $x \neq y^{-1}$ , then the subgroup of  $S_4$  generated by  $x$  and  $y$  is  $A_4$ .

16. Let  $x$  and  $y$  be distinct 3-cycles in  $S_5$  with  $x \neq y^{-1}$ .
- (a) Prove that if  $x$  and  $y$  fix a common element of  $\{1, \dots, 5\}$ , then  $\langle x, y \rangle \cong A_4$ .
  - (b) Prove that if  $x$  and  $y$  do not fix a common element of  $\{1, \dots, 5\}$ , then  $\langle x, y \rangle = A_5$ .
17. If  $x$  and  $y$  are 3-cycles in  $S_n$ , prove that  $\langle x, y \rangle$  is isomorphic to  $Z_3$ ,  $A_4$ ,  $A_5$  or  $Z_3 \times Z_3$ .

# Direct and Semidirect Products and Abelian Groups

1. Show that the center of a direct product is the direct product of the centers:

$$Z(G_1 \times G_2 \times \cdots \times G_n) = Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n).$$

Deduce that a direct product of groups is abelian if and only if each of the factors is abelian.

2. Let  $G_1, G_2, \dots, G_n$  be groups and let  $G = G_1 \times \cdots \times G_n$ . Let  $I$  be a proper, nonempty subset of  $\{1, \dots, n\}$  and let  $J = \{1, \dots, n\} - I$ . Define  $G_I$  to be the set of elements of  $G$  that have the identity of  $G_j$  in position  $j$  for all  $j \in J$ .

(a) Prove that  $G_I$  is isomorphic to the direct product of the groups  $G_i, i \in I$ .

(b) Prove that  $G_I$  is a normal subgroup of  $G$  and  $G/G_I \cong G_J$ .

(c) Prove that  $G \cong G_I \times G_J$ .

3. Under the notation of the preceding exercise let  $I$  and  $K$  be any disjoint nonempty subsets of  $\{1, 2, \dots, n\}$  and let  $G_I$  and  $G_K$  be the subgroups of  $G$  defined above. Prove that  $xy = yx$  for all  $x \in G_I$  and all  $y \in G_K$ .

$$1. (g_1, \dots, g_n) \in Z(\prod G_i) \Leftrightarrow \text{for any } (a_1, \dots, a_n) \in \prod G_i, (g_i) \cdot (a_i) = (a_i) \cdot (g_i) \\ \Leftrightarrow g_i \in Z(G_i) \text{ for each } 1 \leq i \leq n.$$

$$2. G_I = \{(g_1, \dots, g_n) \in G \mid g_i \in G_i, g_j = 1 \text{ if } j \in J\}$$

$$a). G_I \cong \prod_{i \in I} G_i.$$

proof. Define  $\varphi: G_I \rightarrow \prod_{i \in I} G_i: (g_1, \dots, g_n) \mapsto (g_i)_{i \in I}$ . It is clearly onto homomorphism, and

$$\text{Ker } \varphi = \{(g_1, \dots, g_n) \in G_I \mid g_i = 1 \text{ if } i \in I, g_r \in G_r \text{ for each } 1 \leq r \leq n\}.$$

$$= \{(1, \dots, 1)\}, \text{ trivial kernel,}$$

$$\text{By the first Isomorphism Theorem, } G_I / \text{Ker } \varphi \cong G_I \cong \prod_{i \in I} G_i.$$



4. Let  $A$  and  $B$  be finite groups and let  $p$  be a prime. Prove that any Sylow  $p$ -subgroup of  $A \times B$  is of the form  $P \times Q$ , where  $P \in \text{Syl}_p(A)$  and  $Q \in \text{Syl}_p(B)$ . Prove that  $n_p(A \times B) = n_p(A)n_p(B)$ . Generalize both of these results to a direct product of any finite number of finite groups (so that the number of Sylow  $p$ -subgroups of a direct product is the product of the numbers of Sylow  $p$ -subgroups of the factors).
5. Exhibit a nonnormal subgroup of  $Q_8 \times Z_4$  (note that every subgroup of each factor is normal).
6. Show that all subgroups of  $Q_8 \times E_{2^n}$  are normal.

4,

7. Let  $G_1, G_2, \dots, G_n$  be groups and let  $\pi$  be a fixed element of  $S_n$ . Prove that the map

$$\varphi_\pi : G_1 \times G_2 \times \cdots \times G_n \rightarrow G_{\pi^{-1}(1)} \times G_{\pi^{-1}(2)} \times \cdots \times G_{\pi^{-1}(n)}$$

defined by

$$\varphi_\pi(g_1, g_2, \dots, g_n) = (g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)})$$

is an isomorphism (so that changing the order of the factors in a direct product does not change the isomorphism type).

8. Let  $G_1 = G_2 = \dots = G_n$  and let  $G = G_1 \times \dots \times G_n$ . Under the notation of the preceding exercise show that  $\varphi_\pi \in \text{Aut}(G)$ . Show also that the map  $\pi \mapsto \varphi_\pi$  is an injective homomorphism of  $S_n$  into  $\text{Aut}(G)$ . (In particular,  $\varphi_{\pi_1} \circ \varphi_{\pi_2} = \varphi_{\pi_1\pi_2}$ . It is at this point that the  $\pi^{-1}$ 's in the definition of  $\varphi_\pi$  are needed. The underlying reason for this is because if  $e_i$  is the  $n$ -tuple with 1 in position  $i$  and zeros elsewhere,  $1 \leq i \leq n$ , then  $S_n$  acts on  $\{e_1, \dots, e_n\}$  by  $\pi \cdot e_i = e_{\pi(i)}$ ; this is a left group action. If the  $n$ -tuple  $(g_1, \dots, g_n)$  is represented by  $g_1e_1 + \dots + g_ne_n$ , then this left group action on  $\{e_1, \dots, e_n\}$  extends to a left group action on sums by

$$\pi \cdot (g_1e_1 + g_2e_2 + \dots + g_ne_n) = g_1e_{\pi(1)} + g_2e_{\pi(2)} + \dots + g_ne_{\pi(n)}.$$

The coefficient of  $e_{\pi(i)}$  on the right hand side is  $g_i$ , so the coefficient of  $e_i$  is  $g_{\pi^{-1}(i)}$ . Thus the right hand side may be rewritten as  $g_{\pi^{-1}(1)}e_1 + g_{\pi^{-1}(2)}e_2 + \dots + g_{\pi^{-1}(n)}e_n$ , which is precisely the sum attached to the  $n$ -tuple  $(g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)})$ . In other words, any permutation of the “position vectors”  $e_1, \dots, e_n$  (which fixes their coefficients) is the same as the inverse permutation on the coefficients (fixing the  $e_i$ 's). If one uses  $\pi$ 's in place of  $\pi^{-1}$ 's in the definition of  $\varphi_\pi$  then the map  $\pi \mapsto \varphi_\pi$  is not necessarily a homomorphism — it corresponds to a *right* group action.)

9. Let  $G_i$  be a field  $F$  for all  $i$  and use the preceding exercise to show that the set of  $n \times n$  matrices with one 1 in each row and each column is a subgroup of  $GL_n(F)$  isomorphic to  $S_n$  (these matrices are called *permutation matrices* since they simply permute the standard basis  $e_1, \dots, e_n$  (as above) of the  $n$ -dimensional vector space  $F \times F \times \dots \times F$ ).
10. Let  $p$  be a prime. Let  $A$  and  $B$  be two cyclic groups of order  $p$  with generators  $x$  and  $y$ , respectively. Set  $E = A \times B$  so that  $E$  is the elementary abelian group of order  $p^2$ :  $E_{p^2}$ . Prove that the distinct subgroups of  $E$  of order  $p$  are

$$\langle x \rangle, \quad \langle xy \rangle, \quad \langle xy^2 \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

(note that there are  $p + 1$  of them).

11. Let  $p$  be a prime and let  $n \in \mathbb{Z}^+$ . Find a formula for the number of subgroups of order  $p$  in the elementary abelian group  $E_{p^n}$ .

12. Let  $A$  and  $B$  be groups. Assume  $Z(A)$  contains a subgroup  $Z_1$  and  $Z(B)$  contains a subgroup  $Z_2$  with  $Z_1 \cong Z_2$ . Let this isomorphism be given by the map  $x_i \mapsto y_i$  for all  $x_i \in Z_1$ . A *central product* of  $A$  and  $B$  is a quotient

$$(A \times B)/Z \quad \text{where} \quad Z = \{(x_i, y_i^{-1}) \mid x_i \in Z_1\}$$

and is denoted by  $A * B$  — it is not unique since it depends on  $Z_1$ ,  $Z_2$  and the isomorphism between them. (Think of  $A * B$  as the direct product of  $A$  and  $B$  “collapsed” by identifying each element  $x_i \in Z_1$  with its corresponding element  $y_i \in Z_2$ .)

- (a) Prove that the images of  $A$  and  $B$  in the quotient group  $A * B$  are isomorphic to  $A$  and  $B$ , respectively, and that these images intersect in a central subgroup isomorphic to  $Z_1$ . Find  $|A * B|$ .
- (b) Let  $Z_4 = \langle x \rangle$ . Let  $D_8 = \langle r, s \rangle$  and  $Q_8 = \langle i, j \rangle$  be given by their usual generators and relations. Let  $Z_4 * D_8$  be the central product of  $Z_4$  and  $D_8$  which identifies  $x^2$  and  $r^2$  (i.e.,  $Z_1 = \langle x^2 \rangle$ ,  $Z_2 = \langle r^2 \rangle$  and the isomorphism is  $x^2 \mapsto r^2$ ) and let  $Z_4 * Q_8$  be the central product of  $Z_4$  and  $Q_8$  which identifies  $x^2$  and  $-1$ . Prove that  $Z_4 * D_8 \cong Z_4 * Q_8$ .

- 13.** Give presentations for the groups  $Z_4 * D_8$  and  $Z_4 * Q_8$  constructed in the preceding exercise.
- 14.** Let  $G = A_1 \times A_2 \times \cdots \times A_n$  and for each  $i$  let  $B_i$  be a normal subgroup of  $A_i$ . Prove that  $B_1 \times B_2 \times \cdots \times B_n \trianglelefteq G$  and that
- $$(A_1 \times A_2 \times \cdots \times A_n) / (B_1 \times B_2 \times \cdots \times B_n) \cong (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n).$$

5. Let  $G$  be a finite abelian group of type  $(n_1, n_2, \dots, n_t)$ . Prove that  $G$  contains an element of order  $m$  if and only if  $m \mid n_1$ . Deduce that  $G$  is of exponent  $n_1$ .
6. Prove that any finite group has a finite exponent. Give an example of an infinite group with finite exponent. Does a finite group of exponent  $m$  always contain an element of order  $m$ ?
7. Let  $p$  be a prime and let  $A = \langle x_1 \rangle \times \langle x_2 \rangle \times \cdots \times \langle x_n \rangle$  be an abelian  $p$ -group, where  $|x_i| = p^{\alpha_i} > 1$  for all  $i$ . Define the  $p^{\text{th}}$ -power map

$$\varphi : A \rightarrow A \quad \text{by} \quad \varphi : x \mapsto x^p.$$

- (a) Prove that  $\varphi$  is a homomorphism.
- (b) Describe the image and kernel of  $\varphi$  in terms of the given generators.
- (c) Prove both  $\ker \varphi$  and  $A/\text{im } \varphi$  have rank  $n$  (i.e., have the same rank as  $A$ ) and prove these groups are both isomorphic to the elementary abelian group,  $E_{p^n}$ , of order  $p^n$ .

8. Let  $A$  be a finite abelian group (written multiplicatively) and let  $p$  be a prime. Let

$$A^p = \{a^p \mid a \in A\} \quad \text{and} \quad A_p = \{x \mid x^p = 1\}$$

(so  $A^p$  and  $A_p$  are the image and kernel of the  $p^{\text{th}}$ -power map, respectively).

- (a) Prove that  $A/A^p \cong A_p$ . [Show that they are both elementary abelian and they have the same order.]
- (b) Prove that the number of subgroups of  $A$  of order  $p$  equals the number of subgroups of  $A$  of index  $p$ . [Reduce to the case where  $A$  is an elementary abelian  $p$ -group.]
9. Let  $A = \mathbb{Z}_{60} \times \mathbb{Z}_{45} \times \mathbb{Z}_{12} \times \mathbb{Z}_{36}$ . Find the number of elements of order 2 and the number of subgroups of index 2 in  $A$ .



- 10.** Let  $n$  and  $k$  be positive integers and let  $A$  be the free abelian group of rank  $n$  (written additively). Prove that  $A/kA$  is isomorphic to the direct product of  $n$  copies of  $\mathbb{Z}/k\mathbb{Z}$  (here  $kA = \{ka \mid a \in A\}$ ). [See Exercise 14, Section 1.]
- 11.** Let  $G$  be a nontrivial finite abelian group of rank  $t$ .
- (a) Prove that the rank of  $G$  equals the maximum of the ranks of its Sylow subgroups.
  - (b) Prove that  $G$  can be generated by  $t$  elements but no subset with fewer than  $t$  elements generates  $G$ . [One way of doing this is by using part (a) together with Exercise 7.]

- 13.** Let  $A = \langle x_1 \rangle \times \cdots \times \langle x_r \rangle$  be a finite abelian group with  $|x_i| = n_i$  for  $1 \leq i \leq r$ . Find a presentation for  $A$ . Prove that if  $G$  is any group containing commuting elements  $g_1, \dots, g_r$  such that  $g_i^{n_i} = 1$  for  $1 \leq i \leq r$ , then there is a unique homomorphism from  $A$  to  $G$  which sends  $x_i$  to  $g_i$  for all  $i$ .

**14.** For any group  $G$  define the *dual group* of  $G$  (denoted  $\widehat{G}$ ) to be the set of all homomorphisms from  $G$  into the multiplicative group of roots of unity in  $\mathbb{C}$ . Define a group operation in  $\widehat{G}$  by pointwise multiplication of functions: if  $\chi, \psi$  are homomorphisms from  $G$  into the group of roots of unity then  $\chi\psi$  is the homomorphism given by  $(\chi\psi)(g) = \chi(g)\psi(g)$  for all  $g \in G$ , where the latter multiplication takes place in  $\mathbb{C}$ .

(a) Show that this operation on  $\widehat{G}$  makes  $\widehat{G}$  into an abelian group. [Show that the identity is the map  $g \mapsto 1$  for all  $g \in G$  and the inverse of  $\chi \in \widehat{G}$  is the map  $g \mapsto \chi(g)^{-1}$ .]

(b) If  $G$  is a finite abelian group, prove that  $\widehat{\widehat{G}} \cong G$ . [Write  $G$  as  $\langle x_1 \rangle \times \cdots \times \langle x_r \rangle$  and if  $n_i = |x_i|$  define  $\chi_i$  to be the homomorphism which sends  $x_i$  to  $e^{2\pi i/n_i}$  and sends  $x_j$  to 1, for all  $j \neq i$ . Prove  $\chi_i$  has order  $n_i$  in  $\widehat{G}$  and  $\widehat{G} = \langle \chi_1 \rangle \times \cdots \times \langle \chi_r \rangle$ .]

(This result is often phrased: a finite abelian group is self-dual. It implies that the lattice diagram of a finite abelian group is the same when it is turned upside down. Note however that there is no *natural* isomorphism between  $G$  and its dual (the isomorphism depends on a choice of a set of generators for  $G$ ). This is frequently stated in the form: a finite abelian group is *noncanonically* isomorphic to its dual.)

